Project Report Documentation

Date: March 7, 2026

1. PROJECT TITLE

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau and Flask Integration.

2. PROJECT DESCRIPTION

System Overview: An end-to-end data analytics and web integration project that visual- multidimensional retail data to optimize store layouts and product placement.
Problem Domain: Retail Merchandising and Spatial Analytics.
Core Technology Used: Tableau Desktop/Public, Python, Flask Web Framework, HTML/CSS. Key **Capabilities**: Interactive dashboards, cross-filtering (Demographics, Traffic, Promo- tion), data-driven storytelling, and seamless web embedding.
Target Users: Retail Store Managers, Category Managers, and Merchandising Analysts.
Real-world Relevance: Solves the complex issue of manual floor mapping by providing actionable, visual insights into how product positioning (Aisle vs. End-cap) impacts sales volume.

3. APPLICATION SCENARIOS / USE CASES

- Enterprise/Managerial Usage: A store manager preparing for a seasonal rollout uses the embedded dashboard to determine if ”Front of Store” placements yield higher ROI for seasonal clothing than standard aisles.
- Analytical Usage: A merchandising analyst evaluates competitive pricing by filtering the ”Price vs. Competitor Price” scatter plot to identify underperforming electronics categories.

4. TECHNICAL ARCHITECTURE OVERVIEW

Data / Request Flow:

1. Database/Storage Layer: Raw retail sales data is stored in Product Positioning.csv.
2. Analytics Layer: Tableau Desktop connects to the dataset, processes calculated fields, and generates responsive visualizations, dashboards, and stories.
3. Cloud Layer: The Tableau workbook is published to Tableau Public/Cloud to gener- ate an embed code.

4. Backend Web Layer: A Python Flask application (app.py) serves as the host platform, routing user requests.
5. Frontend Layer: HTML/CSS templates render the user interface and execute the Tableau JavaScript API to embed the interactive dashboard directly into the web page.

5. PREREQUISITES

5.1 Software Requirements

- Development Tools: Tableau Desktop 2023+, VS Code or PyCharm (for web dev).
- Runtime Environment: Python 3.8+, modern web browser (Chrome, Edge, Firefox).

5.2 Libraries / Frameworks

- Flask (Python web framework)
- Tableau JavaScript API (Frontend embedding)

6. PRIOR KNOWLEDGE REQUIRED

- Data Analytics: Understanding of basic retail KPIs (Sales Volume, Promotion Lift).
- Tableau Basics: Connecting to data sources, creating worksheets, dashboards, and stories.
- Web Development: Basic understanding of Python (Flask routing) and HTML/iframes.

7. PROJECT OBJECTIVES

- **Technical:** Clean and extract insights from a multi-variable dataset (Product Positioning.csv).
  - Performance: Ensure dashboard load times are minimal and interactive filters re- spond instantly.
  - Deployment: Successfully embed a responsive Tableau dashboard and Story into a custom Flask UI.

8. SYSTEM WORKFLOW

1. User launches the local Flask application via browser (http://127.0.0.1:5000).
2. The Flask route / renders the index.html template.
3. The browser displays a unified dashboard UI featuring a customized sidebar menu, interactive KPI summary cards, and embedded Tableau iframes.
4. User clicks ”Placement Dashboard” or ”Data Story” in the sidebar navigation.

5. A custom JavaScript function (showSection) toggles the active view and updates the header title, fetching the respective interactive visualizations from Tableau Public servers without reloading the page.
6. User interacts with filters (e.g., foot traffic, demographics), dynamically updating all linked charts.

9. MILESTONE 1: REQUIREMENT ANALYSIS

Objective: Data Collection & Extraction.

- Collect the dataset: Sourced Product Positioning.csv containing columns like Prod- uct ID, Position, Price, Foot Traffic, and Sales Volume.
- Connect data with Tableau: Established a live/extract connection to the CSV file within Tableau, ensuring correct data types (e.g., assigning geographic/categorical roles).

10. MILESTONE 2: SYSTEM DESIGN

- Dashboard Layout: Designed a responsive grid featuring a top KPI banner, a left- hand filter pane, and central visualization workspaces.
- Web UI Design: Drafted the Flask HTML templates with Bootstrap for a clean, pro- fessional wrapper around the Tableau iframe.

11. MILESTONE 3: CORE IMPLEMENTATION

11.1 Data Preparation

- Standardized data formats (e.g., converting ”Price” to Currency).
- Created calculated fields, such as [Price Variance] = [Price] - [Competitor's Price].

11.2 Data Visualizations

Developed 4 Unique Visualizations:

1. Bar Chart: Sales Volume grouped by Product Position and Foot Traffic.
2. Scatter Plot: Price vs. Competitor Price categorized by Promotion status.
3. Heatmap: Consumer Demographics vs. Product Category matrix.
4. KPI Text Marks: Aggregate average sales indicators.

11.3 Dashboard & Story Creation

- Dashboard: Combined the 4 unique worksheets into a single, cohesive, responsive Dashboard. Configured highlight actions and global filters.
- Story: Created a 3-Scene Story to guide users through the data: Scene 1 (General Overview), Scene 2 (Impact of Promotions), Scene 3 (Demographic Deep-Dive).

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

Web Integration (Flask):

- Created app.py with routing logic (/ rendering the main index.html template).
- Developed a responsive HTML frontend using Bootstrap 5, featuring a sidebar navigation and stylized KPI cards for quick metrics (Total Products, Traffic Zones, Active Promotions, Avg. Sales).
- Embedded two specific Tableau Public URLs via <iframe> tags: one for the interactive Dashboard and one for the Data Story.
- Implemented custom JavaScript (showSection(sectionId)) to dynamically toggle between the Dashboard and Story views, mapping active CSS classes for a seamless User Experience.

13. MILESTONE 5: TESTING & VALIDATION

13.1 Performance Testing Metrics

- Utilization of Data Filters: Tested global demographic and seasonal filters. Filter execution time < 1 second.
- No. of Calculation Fields: 3 active calculation fields executing flawlessly upon data refresh.
- No. of Visualizations/Graphs: 4 primary graphs render simultaneously without lay- out breaking.

13.2 Test Cases

Test ID	Input / Action	Expected Output	Status
TC-01	Select ”High” in Foot Traffic filter on Dashboard.	All charts dynamically update to show only high-traffic data.	Pass
TC-02	Launch Flask app and navigate to /dashboard.	Webpage loads with embedded Tableau UI visible and interactive.	Pass
TC-03	Click ”Next” on Tableau Story view.	Transitions smoothly to Scene 2 (Impact of Promotions).	Pass

14. DEPLOYMENT

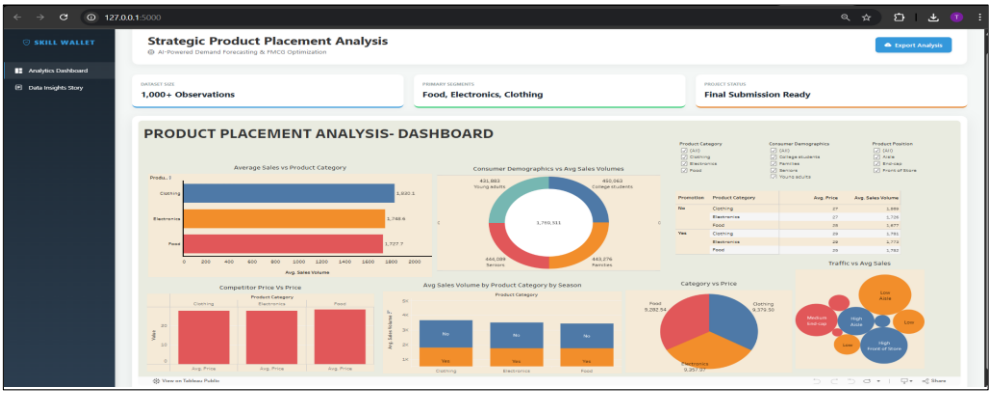
- Deployment Architecture: Local Flask Web Server + Cloud Hosted Tableau Dash- board.
- Hosting Platform: Tableau Public (for visualization hosting) and local environment (for Flask web app).
- Deployment Steps: 1. Publish Workbook to Tableau Public. 2. Copy Embed Code. 3. Insert into Flask index.html. 4. Run python app.py.

15. PROJECT STRUCTURE

```
strategic_placement_project/
|-- data/
|    |-- Product Positioning.csv        # Raw dataset
|-- tableau/
|    |-- Tableau1.twb                   # Core Tableau Workbook
|-- templates/
|    |-- index.html                     # Unified UI with sidebar & embedded Tableau iframes
|-- app.py                              # Main Flask application file routing to index.html
|-- requirements.txt                    # Python dependencies (Flask)
```

16. RESULTS

The system successfully provides a unified web portal where retail managers can view multidimensional positioning data without opening specialized software.



17. ADVANTAGES & LIMITATIONS

- Advantages: Centralized access via a web browser; intuitive visual correlations; no-code interactivity for end-users.
- Limitations: The current architecture relies on a static CSV extract. If point-of-sale data changes, the Tableau extract must be manually refreshed and republished.

18. FUTURE ENHANCEMENTS

- Live Database Integration: Connect Tableau directly to a live SQL database (e.g., Snowflake) instead of a CSV for real-time updates.
- Advanced Flask Integration: Implement Flask-Login to create secure user authentication before viewing the dashboard.

19. CONCLUSION

This project successfully integrates the analytical power of Tableau with the flexibility of a Python Flask web application. By completing the end-to-end process—from raw data collection and preparation to complex visualization, story building, and final web deployment—the solution directly addresses the business need for optimized, data-driven retail merchandise placement.

20. APPENDIX

- Project Documentation: Step-by-step project development procedures are stored in the project's README.md.
- Demonstration Video: *[Insert link to recorded end-to-end solution explanation video here]*
- GitHub Repository Link: *[Insert GitHub URL here]*
- Live Dashboard Link: *[Insert Tableau Public URL here]*

1.1 Source Code: Flask Backend (app.py)

```
@app.route('/')
def home():
    # This renders the HTML page containing your dashboard
    return render_template('index.html')

if __name__ == '__main__':
    # Runs the server in debug mode so you can see changes instantly
    app.run(debug=True, port=5000)
```

Source Code: Frontend UI (index.html)

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Strategic Product Placement | Skill Wallet</title>

    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap" rel="stylesheet">
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.5/font/bootstrap-icons.css">

    <style>
        :root {
            --primary-color: #0d6efd;
            --sidebar-bg: #111827;
            --sidebar-hover: #1f2937;
            --bg-color: #f3f4f6;
            --text-main: #1f2937;
            --text-muted: #6b7280;
        }

        body {
            font-family: 'Inter', sans-serif;
            background-color: var(--bg-color);
            overflow-x: hidden;
            color: var(--text-main);
        }

        /* Sidebar */
        .sidebar {
            height: 100vh;
            background-color: var(--sidebar-bg);
            color: white;
            padding-top: 24px;
            position: fixed;
            width: 260px;
            z-index: 100;
        }

        .brand-logo {
            display: flex;
            align-items: center;
            padding: 0 24px;
            margin-bottom: 2rem;
        }

        .brand-logo i {
            font-size: 1.8rem;
            color: #3b82f6;
            margin-right: 12px;
        }

        .brand-logo div {
            line-height: 1.2;
        }

        .brand-title {
            font-weight: 700;
```

```
            font-size: 1.1rem;
            letter-spacing: 0.5px;
        }

        .brand-subtitle {
            font-size: 0.75rem;
            color: #9ca3af;
            text-transform: uppercase;
            letter-spacing: 1px;
        }

        .nav-label {
            font-size: 0.75rem;
            color: #6b7280;
            text-transform: uppercase;
            letter-spacing: 1px;
            margin: 1.5rem 24px 0.5rem;
            font-weight: 600;
        }

        .sidebar a {
            color: #d1d5db;
            text-decoration: none;
            padding: 12px 24px;
            display: flex;
            align-items: center;
            font-size: 0.95rem;
            font-weight: 500;
            transition: all 0.2s ease;
            cursor: pointer;
            border-left: 3px solid transparent;
        }

        .sidebar a i {
            font-size: 1.1rem;
            margin-right: 12px;
            color: #9ca3af;
        }

        .sidebar a:hover {
            background-color: var(--sidebar-hover);
            color: white;
        }

        .sidebar a:hover i {
            color: white;
        }

        .sidebar a.active {
            background-color: rgba(59, 130, 246, 0.1);
```

```
            color: #3b82f6;
            border-left: 3px solid #3b82f6;
        }

        .sidebar a.active i {
            color: #3b82f6;
        }

        /* Main Content */
        .main-content {
            margin-left: 260px;
            display: flex;
            flex-direction: column;
            min-height: 100vh;
        }

        /* Top Navbar */
        .top-navbar {
            background-color: white;
            height: 70px;
            display: flex;
            align-items: center;
            justify-content: space-between;
            padding: 0 40px;
            border-bottom: 1px solid #e5e7eb;
        }

        .search-bar {
            background-color: #f3f4f6;
            border-radius: 8px;
            padding: 8px 16px;
            display: flex;
            align-items: center;
            width: 300px;
        }

        .search-bar input {
            border: none;
            background: transparent;
            outline: none;
            margin-left: 8px;
            width: 100%;
            font-size: 0.9rem;
        }

        .user-profile {
            display: flex;
            align-items: center;
            gap: 12px;
            font-weight: 500;
```

```
169.            font-size: 0.9rem;
170.        }
171.
172.        .avatar {
173.            width: 36px;
174.            height: 36px;
175.            background-color: #3b82f6;
176.            color: white;
177.            border-radius: 50%;
178.            display: flex;
179.            align-items: center;
180.            justify-content: center;
181.            font-weight: 600;
182.        }
183.
184.        /* Page Content */
185.        .page-content {
186.            padding: 30px 40px;
187.            flex-grow: 1;
188.        }
189.
190.        .badge-skillwallet {
191.            background-color: #e0e7ff;
192.            color: #4338ca;
193.            padding: 4px 10px;
194.            border-radius: 20px;
195.            font-size: 0.75rem;
196.            font-weight: 600;
197.            margin-bottom: 10px;
198.            display: inline-block;
199.        }
200.
201.        h2#main-title {
202.            font-weight: 700;
203.            font-size: 1.75rem;
204.            margin-bottom: 4px;
205.            color: #111827;
206.        }
207.
208.        /* KPI Cards */
209.        .kpi-card {
210.            background: white;
211.            border-radius: 12px;
212.            padding: 20px;
213.            border: 1px solid #e5e7eb;
214.            box-shadow: 0 1px 3px rgba(0,0,0,0.05);
215.            display: flex;
216.            align-items: center;
217.            gap: 16px;
218.            transition: transform 0.2s ease;
```

```
        }

        .kpi-card:hover {
            transform: translateY(-2px);
            box-shadow: 0 4px 6px rgba(0,0,0,0.05);
        }

        .kpi-icon {
            width: 48px;
            height: 48px;
            border-radius: 10px;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 1.4rem;
        }

        .kpi-icon.blue { background: #eff6ff; color: #3b82f6; }
        .kpi-icon.green { background: #f0fdf4; color: #22c55e; }
        .kpi-icon.purple { background: #faf5ff; color: #a855f7; }
        .kpi-icon.orange { background: #fff7ed; color: #f97316; }

        .kpi-details p {
            margin: 0;
            font-size: 0.85rem;
            color: var(--text-muted);
            font-weight: 500;
        }

        .kpi-details h4 {
            margin: 0;
            font-weight: 700;
            font-size: 1.25rem;
            color: #111827;
        }

        /* Dashboard Container */
        .dashboard-wrapper {
            background: white;
            border-radius: 12px;
            padding: 0;
            box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -1px rgba(0, 0, 0, 0.03);
            border: 1px solid #e5e7eb;
            min-height: 850px;
            overflow: hidden;
            margin-top: 24px;
        }

        .view-section {
```

```
            display: none;
            animation: fadeIn 0.3s ease-in-out;
        }

        .view-section.active {
            display: block;
        }

        @keyframes fadeIn {
            from { opacity: 0; transform: translateY(5px); }
            to { opacity: 1; transform: translateY(0); }
        }

        .btn-primary {
            background-color: #3b82f6;
            border-color: #3b82f6;
            font-weight: 500;
            padding: 8px 20px;
            border-radius: 6px;
        }

        .btn-primary:hover {
            background-color: #2563eb;
            border-color: #2563eb;
        }

    </style>
</head>

<body>

    <div class="sidebar">
        <div class="brand-logo">
            <i class="bi bi-wallet2"></i>
            <div>
                <div class="brand-title">Skill Wallet</div>
                <div class="brand-subtitle">Mini Project</div>
            </div>
        </div>

        <div class="nav-label">Analytics Menu</div>
        <a onclick="showSection('dashboard')" id="nav-dashboard" class="active">
            <i class="bi bi-grid-1x2-fill"></i>
            Placement Dashboard
        </a>

        <a onclick="showSection('story')" id="nav-story">
            <i class="bi bi-file-earmark-bar-graph-fill"></i>
            Data Story
        </a>
```

```
    </div>

    <div class="main-content">

        <div class="top-navbar">
            <div class="search-bar">
                <i class="bi bi-search text-muted"></i>
                <input type="text" placeholder="Search insights...">
            </div>
            <div class="user-profile">
                <i class="bi bi-bell text-muted me-3" style="font-size: 1.2rem; cursor: pointer;"></i>
                <div class="avatar">DA</div>
                <span>Data Analyst</span>
            </div>
        </div>

        <div class="page-content">
            <div class="d-flex justify-content-between align-items-end mb-4">
                <div>
                    <span class="badge-skillwallet"><i class="bi bi-check-circle-fill me-1"></i> Skill Wallet Project</span>
                    <h2 id="main-title">Strategic Product Placement Analysis</h2>
                    <p class="text-muted mb-0">
                        Analyzing the impact of store positions on sales volume and demographics.
                    </p>
                </div>
                <div>
                    <button class="btn btn-primary">
                        <i class="bi bi-download me-2"></i>Export Report
                    </button>
                </div>
            </div>

            <div class="row g-4 mb-2">
                <div class="col-md-3">
                    <div class="kpi-card">
                        <div class="kpi-icon blue"><i class="bi bi-box-seam"></i></div>
                        <div class="kpi-details">
                            <p>Total Products</p>
                            <h4>1,000</h4>
                        </div>
                    </div>
                </div>
                <div class="col-md-3">
                    <div class="kpi-card">
                        <div class="kpi-icon green"><i class="bi bi-shop"></i></div>
                        <div class="kpi-details">
                            <p>Highest Traffic Zone</p>
```

```
                            <h4>Front of Store</h4>
                        </div>
                    </div>
                </div>
                <div class="col-md-3">
                    <div class="kpi-card">
                        <div class="kpi-icon purple"><i class="bi bi-tags"></i></div>
                        <div class="kpi-details">
                            <p>Active Promotions</p>
                            <h4>48%</h4>
                        </div>
                    </div>
                </div>
                <div class="col-md-3">
                    <div class="kpi-card">
                        <div class="kpi-icon orange"><i class="bi bi-graph-up-arrow"></i></div>
                        <div class="kpi-details">
                            <p>Avg. Sales Volume</p>
                            <h4>1,984</h4>
                        </div>
                    </div>
                </div>
            </div>

            <div class="dashboard-wrapper">
                <div id="dashboard" class="view-section active">
                    <iframe
                        src="https://public.tableau.com/views/Book1_17727354828350/Dashboard1?:showVizHome=no&:embed=true"
                        width="100%"
                        height="850"
                        style="border:none;">
                    </iframe>
                </div>

                <div id="story" class="view-section">
                    <iframe
                        src="https://public.tableau.com/views/Book1_17727354828350/Story1?:showVizHome=no&:embed=true"
                        width="100%"
                        height="850"
                        style="border:none;">
                    </iframe>
                </div>
            </div>

        </div>
    </div>

```

```
    <script>
        function showSection(sectionId) {
            // Hide all sections
            document.querySelectorAll('.view-section').forEach(section => {
                section.classList.remove('active');
            });

            // Show selected section
            document.getElementById(sectionId).classList.add('active');

            // Remove active class from all links
            document.querySelectorAll('.sidebar a').forEach(link => {
                link.classList.remove('active');
            });

            // Add active class to clicked link
            document.getElementById('nav-' + sectionId).classList.add('active');

            // Update Header Title based on selection
            const title = document.getElementById('main-title');
            title.innerText =
                sectionId === 'dashboard'
                    ? 'Strategic Product Placement Analysis'
                    : 'FMCG Strategic Data Story';
        }
    </script>

    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```